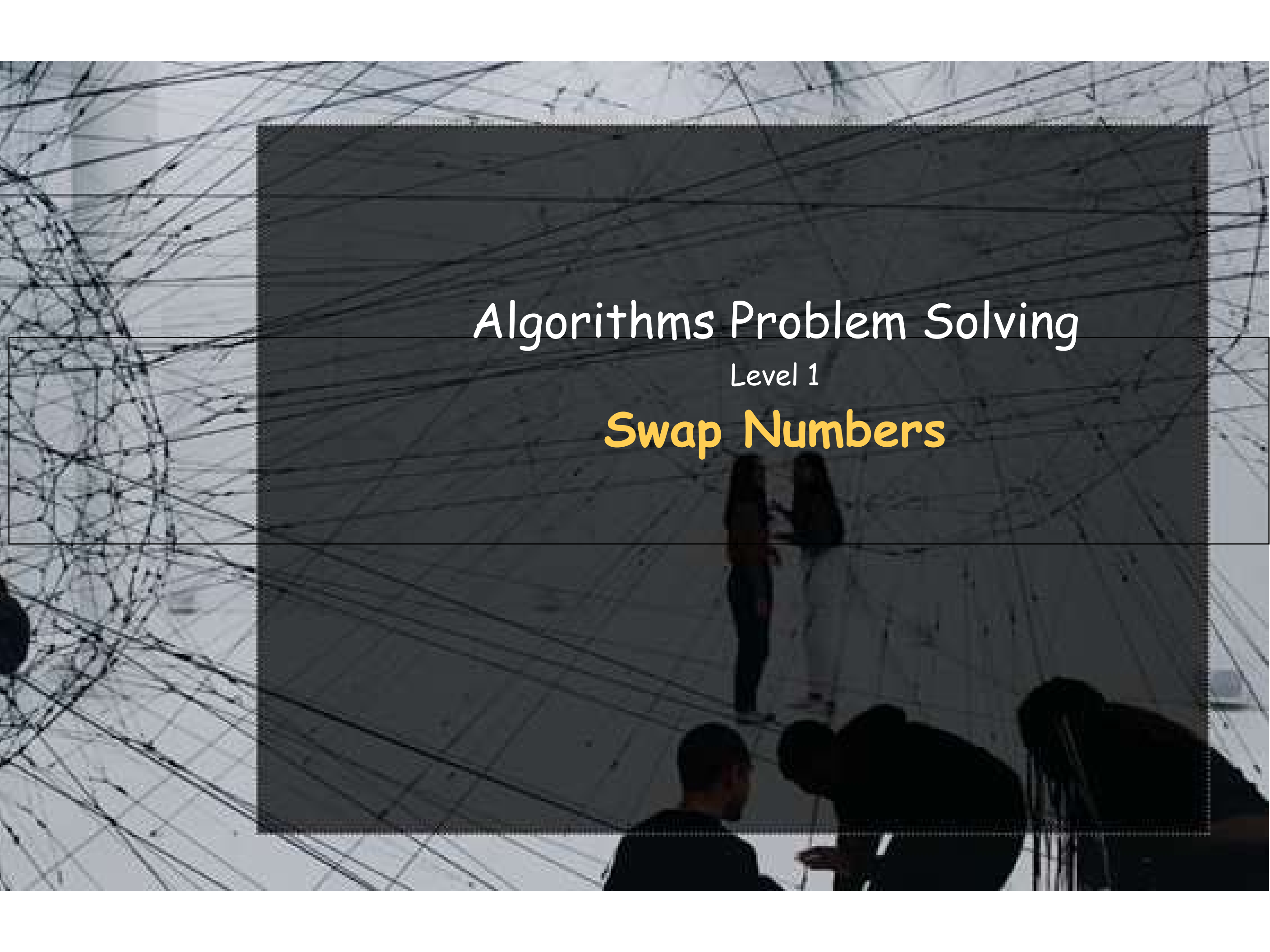


Algorithms Problem Solving

Level 1

Swap Numbers



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Definition

- ◆ This program reads two numbers, displays them, swaps their values, and then displays them again.
- ◆ The program uses functions, reference parameters, and a temporary variable to swap the values.

Explanation

- ◆ The `readNumbers()` function reads two numbers and stores them in `num1` and `num2`.
- ◆ The parameters `int& num1` and `int& num2` are reference parameters, so the function can modify the original variables from `main()`.
- ◆ The `swapNumbers()` function exchanges the values of the two variables:
- ◆ `Temp` stores the original value of `num1`.
- ◆ `num1` receives the value of `num2`.
- ◆ `num2` receives the original value stored in `Temp`.
- ◆ The `printNumbersBefore()` function displays the numbers before swapping.
- ◆ The `printNumbersAfter()` function displays the numbers after swapping.
- ◆ In `main()`, the functions are called in order:
- ◆ `readNumbers(Num1, Num2);`
- ◆ `printNumbersBefore(Num1, Num2);`
- ◆ `swapNumbers(Num1, Num2);`
- ◆ `printNumbersAfter(Num1, Num2);`

Steps in main()

1. Create two integer variables.
2. Read the two numbers.
3. Display the numbers before swapping.
4. Store num1 in a temporary variable.
5. Assign num2 to num1.
6. Assign the temporary value to num2.
7. Display the numbers after swapping.

💡 Example

◆ If the user enters:

Number 1: 10

Number 2: 20

◆ Before swapping:

10 , 20

◆ Temp stores 10.

◆ num1 becomes 20.

◆ num2 becomes 10.

➡ Result

After swapping: 20 , 10

Note

- ◆ A temporary variable is needed to keep the original value of num1 while the values are being exchanged.
- ◆ Reference parameters allow swapNumbers() to modify the original variables directly.

⊘ Warning

⚡ If you assign `num1 = num2` before saving the original `num1` value, the original value of `num1` will be lost.



Comparison

▨ Before Swapping

- ✓ num1 = 10.
- ✓ num2 = 20.
- ✓ The values are in their original positions.

▨ After Swapping

- ✓ num1 = 20.
- ✓ num2 = 10.
- ✓ The values have exchanged positions.

✦✦ Summary

- ✓ Swapping means exchanging the values of two variables.
- ✓ A temporary variable can be used to safely exchange values.
- ✓ Reference parameters allow a function to modify the original variables.
- ✓ The order of assignments is important when swapping values.
- ✓ Functions can separate reading, processing, and printing tasks.

Thank you 😊